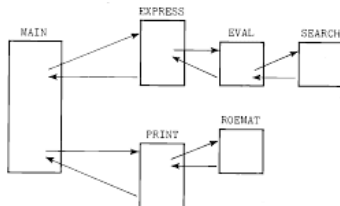


Unidade - 07

Dividindo o programa em blocos.



```
1  ▼ PROGRAM seno_exemplo
2  IMPLICIT NONE
3  REAL(KIND=8):: angle_d,x,angle_rad
4  WRITE(*,*) 'Entre com um angulo em graus.'
5  READ(*,*) angle_d
6  angle_rad=angle_d*3.1416D0/180.0D0
7  x=DSIN(angle_rad)
8  WRITE(*,*) 'O seno de ',angle_d,' vale ',x
9  END PROGRAM seno_exemplo
```

```
1  ▼ PROGRAM cosseno_exemplo
2  IMPLICIT NONE
3  REAL(KIND=8):: angle_d,x,angle_rad
4  WRITE(*,*) 'Entre com um angulo em graus.'
5  READ(*,*) angle_d
6  angle_rad=angle_d*3.1416D0/180.0D0
7  x=DCOS(angle_rad)
8  WRITE(*,*) 'O cosseno de ',angle_d,' vale ',x
9  END PROGRAM cosseno_exemplo
```

Funções Intrínsecas

```
1 ▼ PROGRAM raiz_exemplo
2   IMPLICIT NONE
3   REAL(KIND=8):: x,raiz
4   WRITE(*,*) 'Entre com um valor.'
5   READ(*,*) x
6   raiz=DSQRT(x)
7   WRITE(*,*) 'A raiz quadrada de ',x,' vale ',raiz
8   END PROGRAM raiz_exemplo
```

Funções Intrínsecas

y=**SIN**(x)

y=**DSIN**(x)

y=**COS**(x)

y=**DCOS**(x)

y=**TAN**(x)

y=**DTAN**(x)

y=**SQRT**(x)

y=**DSQRT**(x)

y=**FLOAT**(i)

y=**DFLOAT**(i)

y=**EXP**(x)

y=**DEXP**(x)

y=**ACOS**(x)

y=**DACOS**(x)

y=**ASIN**(x)

y=**DASIN**(x)

y=**AIMAG**(x)

y=**REAL**(x)

y=**CONJG**(x)

y=**ABS**(x)

y=**DABS**(x)

y=**LOG**(x)

y=**DLOG**(x)

y=**LOG10**(x)

y=**TRIM**(a)

y=**ADJUSTL**(a)

y=**ADJUSTR**(a)

y=**FLOOR**(x)

y=**CEILING**(x)

y=**MOD**(i, j)

z=**MIN**(x, y)

z=**MAX**(x, y)

Criando suas próprias Funções

```
1  ▼ FUNCTION seno_exp(x)
2    IMPLICIT NONE
3    REAL(KIND=8) :: x, expo, seno_exp
4    expo=DEXP(x)
5    seno_exp=DSIN(expo)
6    END FUNCTION seno_exp
```

Criando suas próprias Funções

```
1  ▼ FUNCTION cubic_root(x)
2  IMPLICIT NONE
3  REAL(KIND=8):: x, cubic_root
4  cubic_root=x**(1.0D0/3.0D0)
5  END FUNCTION cubic_root
```

Utilizando sua Função

```
1  ▼ PROGRAM calculate_cubic_roots
2  IMPLICIT NONE
3  INTEGER:: n
4  REAL(KIND=8):: nreal,root,cubic_root
5  ▼ DO n=1,20
6      nreal=DFLOAT(n)
7      root=cubic_root(nreal)
8      WRITE(*,*) 'A raiz cubica de ',n,'vale',root
9  END DO
10 END PROGRAM calculate_cubic_roots
```


Arquivo roots.f90

```
1 PROGRAM calculate_cubic_roots
2 IMPLICIT NONE
3 INTEGER:: n
4 REAL(KIND=8):: nreal, root, cubic_root
5 DO n=1,20
6     nreal=DFLOAT(n)
7     root=cubic_root(nreal)
8     WRITE(*,*) 'A raiz cubica de ',n,'vale',root
9 END DO
10 END PROGRAM calculate_cubic_roots
11
12 FUNCTION cubic_root(x)
13 IMPLICIT NONE
14 REAL(KIND=8):: x, cubic_root
15 cubic_root=x**(1.0D0/3.0D0)
16 END FUNCTION cubic_root
```

Arquivo roots.f90

```
1 PROGRAM calculate_cubic_roots
2 IMPLICIT NONE
3 INTEGER:: n
4 REAL(KIND=8):: nreal, root, cubic_root
5 DO n=1,20
6     nreal=DFLOAT(n)
7     root=cubic_root(nreal)
8     WRITE(*,*) 'A raiz cubica de ',n,'vale',root
9 END DO
10 END PROGRAM calculate_cubic_roots
11
12 FUNCTION cubic_root(x)
13 IMPLICIT NONE
14 REAL(KIND=8):: x, cubic_root
15 cubic_root=x**(1.0D0/3.0D0)
16 END FUNCTION cubic_root
```

Tudo em um arquivo só.

Arquivo roots.f90

```
1 PROGRAM calculate_cubic_roots
2 IMPLICIT NONE
3 INTEGER:: n
4 REAL(KIND=8):: nreal, root, cubic_root
5 DO n=1,20
6     nreal=DFLOAT(n)
7     root=cubic_root(nreal)
8     WRITE(*,*) 'A raiz cubica de ',n,' vale ',root
9 END DO
10 END PROGRAM calculate_cubic_roots
11
12 FUNCTION cubic_root(x)
13 IMPLICIT NONE
14 REAL(KIND=8):: x, cubic_root
15 cubic_root=x**(1.0D0/3.0D0)
16 END FUNCTION cubic_root
```

Tudo em um arquivo só.

Para compilar:

gfortran -o roots.x roots.f90

Arquivo roots.f90

```
1 PROGRAM calculate_cubic_roots
2 IMPLICIT NONE
3 INTEGER:: n
4 REAL(KIND=8):: nreal, root, cubic_root
5 DO n=1,20
6     nreal=DFLOAT(n)
7     root=cubic_root(nreal)
8     WRITE(*,*) 'A raiz cubica de ',n,'vale ',root
9 END DO
10 END PROGRAM calculate_cubic_roots
```

Arquivo cubic_root.f90

```
1 FUNCTION cubic_root(x)
2 IMPLICIT NONE
3 REAL(KIND=8):: x, cubic_root
4 cubic_root=x**(1.0D0/3.0D0)
5 END FUNCTION cubic_root
```

Arquivo roots.f90

```
1 PROGRAM calculate_cubic_roots
2 IMPLICIT NONE
3 INTEGER:: n
4 REAL(KIND=8):: nreal, root, cubic_root
5 DO n=1,20
6     nreal=DFLOAT(n)
7     root=cubic_root(nreal)
8     WRITE(*,*) 'A raiz cubica de ',n,'vale ',root
9 END DO
10 END PROGRAM calculate_cubic_roots
```

Arquivo cubic_root.f90

```
1 FUNCTION cubic_root(x)
2 IMPLICIT NONE
3 REAL(KIND=8):: x, cubic_root
4 cubic_root=x**(1.0D0/3.0D0)
5 END FUNCTION cubic_root
```

Programa principal e função em arquivos separados.

Arquivo roots.f90

```
1 PROGRAM calculate_cubic_roots
2 IMPLICIT NONE
3 INTEGER:: n
4 REAL(KIND=8):: nreal, root, cubic_root
5 DO n=1,20
6     nreal=DFLOAT(n)
7     root=cubic_root(nreal)
8     WRITE(*,*) 'A raiz cubica de ',n,'vale ',root
9 END DO
10 END PROGRAM calculate_cubic_roots
```

Arquivo cubic_root.f90

```
1 FUNCTION cubic_root(x)
2 IMPLICIT NONE
3 REAL(KIND=8):: x, cubic_root
4 cubic_root=x**(1.0D0/3.0D0)
5 END FUNCTION cubic_root
```

Programa principal e função em arquivos separados.

Para compilar:

```
gfortran -o roots.x roots.f90 cubic_root.f90
```

Extra em compilação

```
edu@pegaso: ~/programas
File Edit View Search Terminal Help
edu@pegaso:~/programas$ ls
cubic_root.f90  roots.f90
edu@pegaso:~/programas$ gfortran -c cubic_root.f90
edu@pegaso:~/programas$ ls
cubic_root.f90  cubic_root.o  roots.f90
edu@pegaso:~/programas$ gfortran -c roots.f90
edu@pegaso:~/programas$ ls
cubic_root.f90  cubic_root.o  roots.f90  roots.o
edu@pegaso:~/programas$ gfortran -o roots.x roots.o cubic_root.o
edu@pegaso:~/programas$ ls
cubic_root.f90  cubic_root.o  roots.f90  roots.o  roots.x
edu@pegaso:~/programas$
edu@pegaso:~/programas$
edu@pegaso:~/programas$
edu@pegaso:~/programas$ gfortran -c cubic_root.f90
edu@pegaso:~/programas$ ls
cubic_root.f90  cubic_root.o  roots.f90  roots.o  roots.x
edu@pegaso:~/programas$ gfortran -o roots.x roots.o cubic_root.o
edu@pegaso:~/programas$ ls
cubic_root.f90  cubic_root.o  roots.f90  roots.o  roots.x
edu@pegaso:~/programas$
```

PROGRAM prog

.....

.....

.....

.....

.....

%%%%%

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&&&&&

#####

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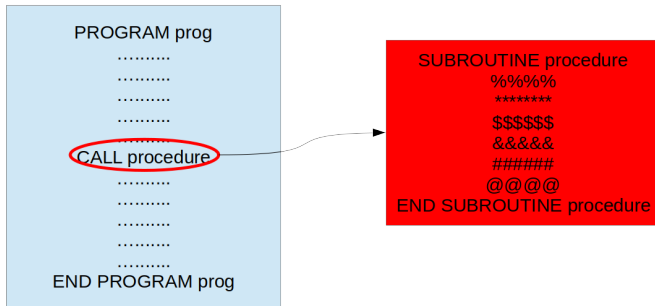
.....

.....

.....

END PROGRAM prog

Subrotinas



```
SUBROUTINE calcula_media_3(x1,x2,x3,media)
```

```
IMPLICIT none
```

```
REAL,INTENT(IN):: x1,x2,x3
```

```
REAL,INTENT(OUT):: media
```

```
media=(x1+x2+x3)/3.0
```

```
END SUBROUTINE calcula_media_3
```

Chamando uma subrotina

```
PROGRAM media_3
```

```
IMPLICIT none
```

```
REAL:: number1,number2,number3,med
```

```
PRINT*, 'Digite o primeiro número'
```

```
READ*, number1
```

```
PRINT*, 'Digite o segundo número'
```

```
READ*, number2
```

```
PRINT*, 'Digite o terceiro número'
```

```
READ*, number3
```

```
CALL calcula_media(number1,number2,number3,med)
```

```
PRINT*, 'A média é igual a', med
```

```
END PROGRAM media_3
```

Ellis book, exercises:

- 4.1
- 4.2
- 4.9
- 4.10
- 4.11
- 4.12